

Learning to Read Where to Look: Disease-Aware Vision-Language Pretraining for 3D CT (Oral)

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Overview

Scan-report alignment is useful, but radiology reports contain more supervision than current 3D CT vision-language models use.

Other works learn global scan-report representations, or add organ-level supervision using segmentation models and LLM-generated region descriptions.

RadFinder combines three supervision signals in one 3D CT model: global report alignment, disease-prompt training, and snippet-to-slice localization.

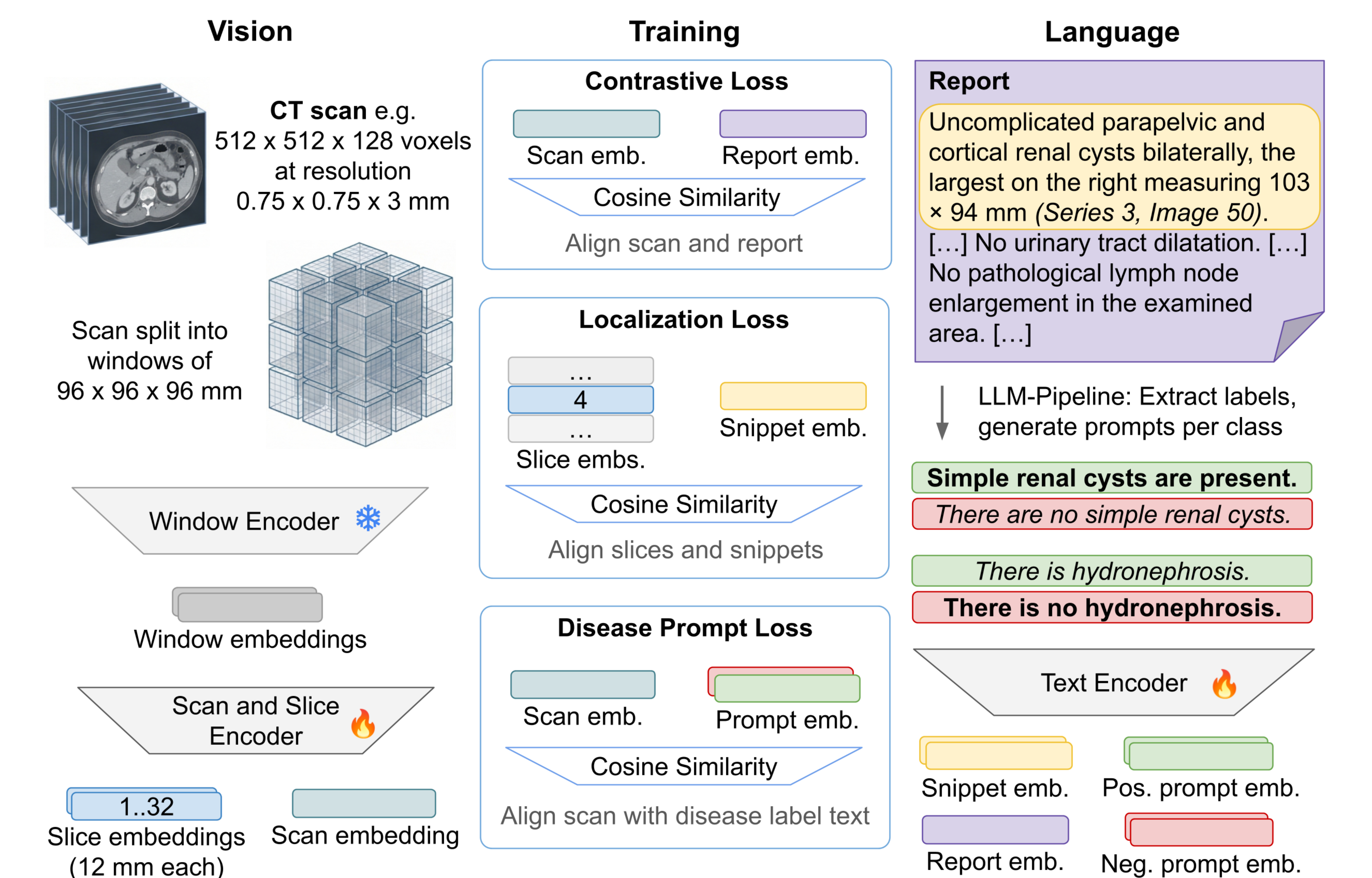
RadFinder adds two report-native signals: disease prompts for classification and slice references for local grounding. This yields one model for retrieval, disease classification, and intra-scan localization.

RefCT dataset

RefCT contains **98k scan-report pairs** from **50k patients**, collected from routine clinical care at Uniklinik Freiburg: chest, abdomen, and combined chest-abdomen studies. **RadFinder** trains on its 78k training split together with the public CT-RATE, Merlin, and INSPECT datasets.

Slice references are extracted automatically from report phrases (snippets) such as "...renal cysts bilaterally, the largest on the right measuring 103 × 94 mm (*Series 3, Image 50*)". In total we mine **262k snippet-slice pairs** that provide local supervision at scale.

RadFinder



Results

Retrieval: Best report-to-scan retrieval on CT-RATE and Merlin. Given a report, RadFinder finds the matching scan in a pool of candidates. Also best scan-to-scan retrieval: scans with similar findings are close in embedding space.

Classification: Competitive disease classification via text prompts on CT-RATE, holding up on the unseen external Rad-ChestCT dataset.

Localization: Predicts the axial position referred to by a short report snippet. RadFinder outperforms all baselines, while retaining retrieval and disease classification performance.

Ablations: Prompt loss alone yields the best classification but sacrifices retrieval. Combined training achieves both. Training on RefCT alone already surpasses all published baselines on CT-RATE retrieval. The contributions of our training setup and our internal data are complementary.

→ A single unified model for all three tasks.

Method	CT-RATE			RC	Merlin
	R@10	MAP@5	AUC	AUC	R@1 _{find}
CT-CLIP	5.0	68.3	73.1	62.9	—
MedVista3D-ViT	10.7	—	77.8	—	—
Merlin	2.7	62.6	72.8	64.4	59.4
BrgSA	22.2	70.4	82.9	74.2	—
MPS-CT	16.3	71.3	83.8	77.3	—
SPECTRE	18.2	76.8	54.3	55.7	44.5
RadFinder (Ours)	31.5	78.0	83.8	77.0	69.0
No localization loss	31.5	78.2	83.6	76.9	69.3
Global loss only	29.4	76.6	56.9	62.8	70.7
Prompt loss only	5.6	77.9	84.8	78.6	13.1
RefCT dataset only	26.3	76.7	80.4	77.7	60.5
Public datasets only	26.9	77.1	83.2	73.6	67.5

Report-to-scan retrieval (R@K), scan-to-scan retrieval (MAP@5), disease classification (AUC) on the CT-RATE, RC (Rad-ChestCT), and Merlin validation sets.

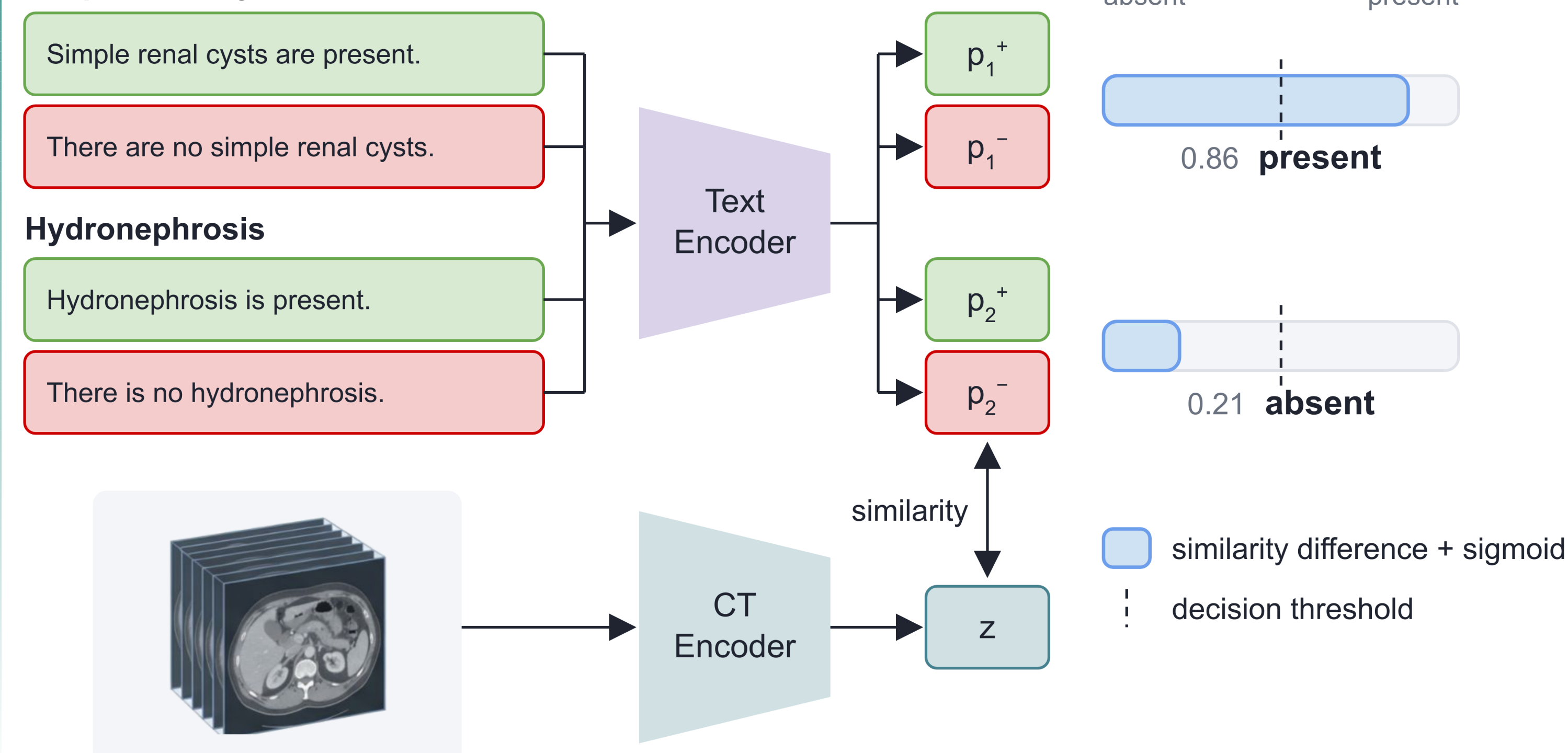
Method	MAE	6 mm	18 mm	30 mm
Random slice	126.9	4.7	11.8	17.8
Middle slice	95.8	4.6	13.0	20.8
CLIP ViT-B/32	124.6	3.2	9.4	14.8
BiomedCLIP	86.6	8.3	20.2	30.5
MedSigLIP-448	75.6	9.7	26.2	40.1
SigLIP2	82.4	7.8	19.3	31.5
SigLIP2 finetuned	67.0	17.4	35.8	48.7
RadFinder	36.3	20.3	45.3	61.8
Loc. loss only	36.0	19.9	45.2	62.7

Snippet localization on the RefCT test set. MAE (mm): Lower is better. Thresholds 6 mm / 18 mm / 30 mm: Percentage of localization errors below this threshold, higher is better. SigLIP2 finetuned uses RefCT snippet-slice pairs with a contrastive objective.

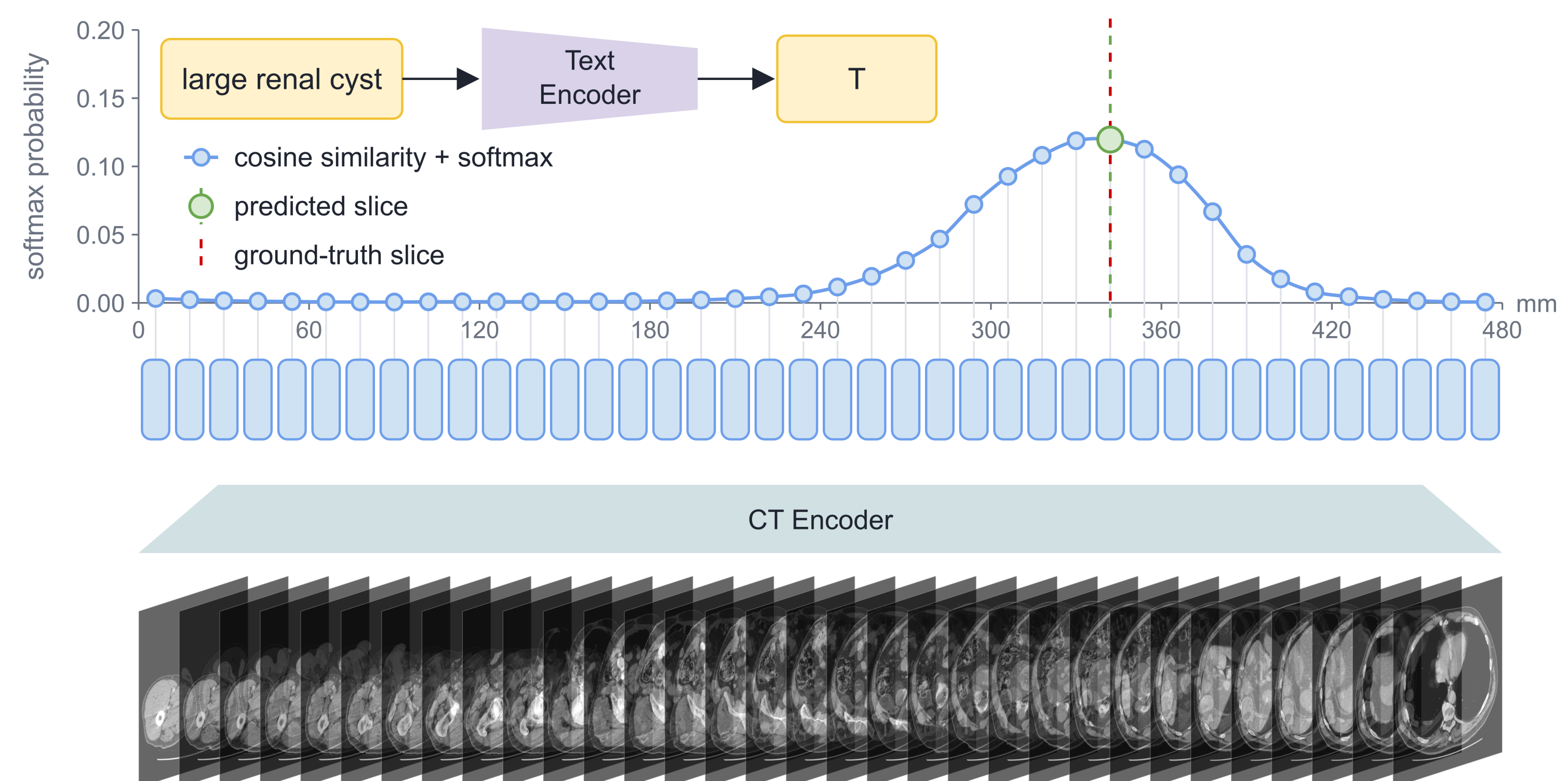
Inference: classification and localization

Disease classification from text prompts. The scan embedding is compared with positive and negative prompt embeddings. The sigmoid of the similarity difference then gives the probability that the finding is present.

Simple renal cysts



Snippet localization at inference. The query embedding is scored against the scan's depth embeddings, one per 12 mm, and the softmax over the axial axis peaks at the predicted slice.



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References. Please refer to our paper for the complete list of references.